

Methodological Convergence in Stage-IV Cosmology: The Emergence of Epistemic Holism Under Precision Constraints, and Its Operationalization via RCMP

Morten Magnusson

Symbiose Research, Sandnes, Norway

ORCID: [0009-0002-4860-5095](https://orcid.org/0009-0002-4860-5095)

Correspondence: morten@magnusson.as

April 2026

DOI: [10.6084/m9.figshare.32080080](https://doi.org/10.6084/m9.figshare.32080080) Status: PUBLISHED v0.2 License: CC-BY-4.0

Abstract

This note documents a methodological trend in cosmological literature from 2023–2026: independent analyses across cosmic shear, gravitational-slip estimators, and dark-energy inference have begun reporting measurement-pipeline limitations with increasing structural explicitness. Specifically, (i) cosmic-shear S_8 tension is now routinely framed as conditional on baryonic feedback and intrinsic-alignment model choice rather than as a model-independent observation; (ii) the E_G gravitational-slip estimator is being re-evaluated through alternative velocity-reconstruction pipelines that bypass GR-assumptive RSD templates; (iii) DESI DR2 dark-energy-evolution hints are reported with explicit supernova-dataset dependence.

These reporting shifts are interpreted here not as field adoption of a new methodology, but as the practical emergence of Duhem–Quine epistemic holism under Stage-IV precision constraints: when measurements are precise enough, previously-implicit proxy-chain assumptions become impossible to hide, and reporting must make them explicit. The Regime-Consistent Measurement Principle (RCMP; Magnusson 2026, DOI: 10.6084/m9.figshare.31222900) offers one concrete operationalization of this constraint, with a 5-point validation checklist and a “problem definition, not resolution” output rule. The convergence observed in the literature is consistent with RCMP-type reasoning becoming necessary in practice, regardless of whether RCMP itself is cited or adopted.

This note claims methodological convergence under Stage-IV pressure, not RCMP endorsement. It does not claim EFC or any specific physical model is supported or disfavoured by this convergence. Quantitative EFC tests are specified in §6 and gated by the release conditions in §9.

The test plan in §6 operates at the MGCAMB/MGCLASS-native phenomenological (μ, Σ) parametrization level, chosen for likelihood-pipeline compatibility. The corresponding action-level EFC parameters (K_0, V', α) are not in one-to-one correspondence with (A_μ, A_Σ) . This action-to-phenomenology mapping is a separate open task flagged in the companion compound paper on $P1 \times P2$ cross-observable consistency.

1 Background

The Regime-Consistent Measurement Principle (RCMP) was published February 2026 as a methodological framework for physical measurements spanning multiple regimes. It specifies five required components for validity: driver-proximity, regime-tagging, proxy-accounting, coordinate-humility, cross-validation. A key distinctive feature is that RCMP-compliant conclusions are formulated as *problem definitions, not resolutions* — that is, they describe what can and cannot be measured under specified conditions, rather than asserting which physical model is preferred.

RCMP is a contemporary operationalization of the Duhem–Quine thesis (Duhem 1906; Quine 1951) for Stage-IV precision cosmology, with an explicit checklist-based workflow. The underlying epistemic insight — that no physical measurement is theory-neutral, and that proxy-chain assumptions are inseparable from the measured quantity — is not original to RCMP; it has been a philosophical commitment of physics since the early 20th century, and has been discussed explicitly in cosmological contexts around H_0 reconstruction debates (Sandage 1988; Freedman et al. 2001), gravitational-slip estimators (Pogosian & Silvestri 2016), and S_8 -tension interpretation (Amon & Efstathiou 2022).

What RCMP contributes is not the discovery of this epistemic situation, but its operationalization: a concrete checklist, a regime-tagging protocol (L0–L3), proxy-chain documentation with propagated uncertainties, and an explicit “problem definition, not resolution” rule for conclusions. RCMP was developed as internal methodology for Energy-Flow Cosmology (EFC; Magnusson 2026, figshare 31112536 and successors) but its structural claims apply to any multi-regime measurement program.

The convergence observed below is therefore not evidence of RCMP’s adoption as framework. It is evidence that Stage-IV precision is forcing pre-existing proxy-chain dependencies into explicit reporting — which RCMP argues was the correct epistemic frame throughout. The specific claim of this note is that Stage-IV data will make RCMP-type reasoning unavoidable in practice, regardless of whether RCMP itself is adopted by name.

2 Observation 1: Cosmic-Shear S_8 Reporting

2.1 Trend

Cosmic-shear analyses of the past three years have increasingly framed S_8 inference as conditional on nuisance-model choice, rather than as a direct measurement of structure growth.

- **Preston, Amon & Efstathiou (2023; MNRAS 525, 5554):** proposed that S_8 tension in DES Y3 is compatible with non-linear matter-power suppression whose required amplitude depends on the baryonic-feedback model adopted. Explicitly frames S_8 as a derived quantity contingent on feedback modeling.
- **Wright et al. (2025), KiDS-Legacy:** S_8 converges toward Planck when intrinsic-alignment modeling is revised; the tension described in earlier KiDS-1000 analysis is reported as “primarily systematics-driven” rather than physics-driven.
- **Abbott et al. (2026), DES Y6 cosmic shear:** reports $2.4\text{--}2.7\sigma$ S_8 -tension with CMB under fiducial NLA intrinsic-alignment model, shifting to 1.7σ under TATT. The sensitivity to IA-choice is reported as a primary uncertainty, not a secondary systematic.
- **2026 S_8 Tension Review (arXiv:2602.12238):** characterizes the field as showing “striking bifurcation” between surveys, with explicit acknowledgment that the bifurcation “complicates the hypothesis of genuine new physics” without resolving it.

2.2 Parallel to RCMP

These reporting choices map directly onto RCMP’s *coordinate-humility* and *proxy-accounting* principles: S_8 is treated as a transformation product of shear measurements through nuisance-marginalized Boltzmann forward models, rather than as a direct observable. The resulting conclusions are phrased as problem definitions (“the sensitivity to IA choice limits what can be stated”) rather than resolutions (“ S_8 from weak lensing is X”).

2.3 What this does and does not imply

It implies: the field is converging toward the kind of structural caveat-reporting RCMP formalizes.

It does not imply: EFC’s P4 prediction ($S_8 = 0.847$ in Rubin DP2) is supported. P4 remains testable in DP2; the convergence observation is orthogonal to that test.

3 Observation 2: E_G Estimator Re-evaluation

3.1 Trend

The E_G gravitational-slip estimator has been the subject of active methodological revision.

- **Wenzl et al. (2024), and the ACT DR6 $\times$ SDSS BOSS analysis:** report $E_G(z=0.555) = 0.31^{+0.06}_{-0.05}$ with 3–4% systematic uncertainty, below the GR expectation (~ 0.40 under Planck-best-fit). The paper explicitly flags systematic uncertainties in the β -reconstruction step.
- **Kamran et al. (2026, arXiv:2510.27605), Phys. Rev. D 113, 023527:** propose a new E_G estimator using kSZ-reconstructed velocities instead of RSD-derived β . Motivation stated: existing E_G measurements inherit systematics from RSD template choices that assume GR-Kaiser form.
- **Hu & Raveri (2025):** demonstrate that the E_G statistic’s GR-invariance property is conditional on both the Poisson equation and the RSD template used in β -extraction; under modified-gravity forward models, the invariance weakens.

3.2 Parallel to RCMP

This corresponds to RCMP’s requirement that proxy chains be documented with their transformation assumptions. The field is moving toward explicit recognition that E_G — long considered a “gold-standard” GR-invariant statistic — carries structural GR-assumptions in its proxy chain (galaxy lensing $\rightarrow$ RSD $\beta \rightarrow E_G$) that break under MG-forward evaluation.

3.3 Implication for EFC P3

EFC’s P3 prediction ($E_G^{\text{EFC}}/E_G^{\text{GR}} = 1.086 \pm 0.012$, L2-regime, pre-registered seal per DOI 10.6084/m9.figshare.32037990) is evaluated at scales where the EFC quasi-static slip is scale-independent per the action-level derivation. The value $k_c = 0.05 \, h/\text{Mpc}$ quoted in downstream MGCAMB parametrizations (Test Spec §1.2, §6.1 below) serves as a phenomenological scale marker, not as the defining scale of the P3 prediction itself.

P3 currently appears “in the wrong direction” relative to ACT DR6 $\times$ BOSS ($E_G(z=0.555) = 0.31^{+0.06}_{-0.05}$ per Wenzl et al. 2024). Under RCMP, the appropriate conclusion is not “P3 is falsified” but “the pipeline does not yet support a coordinate-humble test of P3” — specifically because the β -reconstruction step assumes GR-Kaiser RSD templates that break under EFC-forward evaluation per Hu & Raveri (2025). The Kamran et al. (2026) kSZ-reconstructed E_G estimator offers a GR-assumption-free alternative and is the appropriate target for P3 evaluation in upcoming Stage-IV data. This is a problem-definition, not a defense of EFC.

4 Observation 3: Dark-Energy Evolution Inference

4.1 Trend

DESI DR2 BAO results, combined with various SN samples, have yielded reported preferences for dynamical dark energy (DDE) at $2.8\text{--}4.2\sigma$ — with the statistical significance depending on which SN dataset is combined (Pantheon+, DES-SN5YR, Union3).

- **Wang et al. (2025), European Physical Journal C:** “DESI’s claim of dynamical dark energy is not robust” — explicitly attributes variation to dataset-combination choices, not to new physics per se.
- **Sen et al. (2025, JHEAp 49, 100428):** perform Bayesian reconstruction of $w(z)$ and find “mild, oscillatory departure from a cosmological constant” across all SN subsets, framed as a problem of constraint stability rather than a discovery claim.
- **DESI Collaboration response:** acknowledges in published commentary that the $w_0\text{--}w_a$ parameterization may not adequately capture the preferred solution.

4.2 Parallel to RCMP

This is RCMP’s cross-validation principle operating in practice: the same physical quantity ($w(z)$ evolution) is measured via multiple proxy chains (DESI+Pantheon+, DESI+DES-SN5YR, DESI+Union3), divergence is reported explicitly, and the conclusion is framed as problem (“what can be measured under these parameterizations?”) rather than resolution (“DDE is confirmed/refuted”).

4.3 Implication for EFC P6

EFC’s P6 ($\Delta E^2(z) \leq 0$ for $z > 0$) falls within the class of non-monotonic expansion histories that the DESI data is probing. The RCMP-compliant statement is that P6 is a candidate structural form within the class where data currently points, not that it is supported.

5 What this Convergence Does Not Mean

Three explicit disclaimers.

5.1 The field has not adopted RCMP. No paper cited above references RCMP. The convergence is substantive, not nominal. Mainstream cosmology arrived at caveat-dense reporting through the constraints of Stage-III data, not through adopting RCMP.

5.2 RCMP does not resolve EFC’s empirical obligations. The convergence observation does not reduce the requirement for EFC to deliver quantitative forward-modeled tests against DES Y6 $\xi_{\pm}(\theta)$, KiDS Legacy, ACT DR6, Euclid DR1, etc. See §6 for the test plan that operationalizes this obligation.

5.3 Convergence of reporting style is not validation of EFC model. The field can converge toward RCMP-like caveat-density while simultaneously rejecting EFC on data-specific grounds. These are independent questions.

6 Test Plan: Operationalizing the RCMP Position

The convergence observation has no epistemic weight without parallel delivery of quantitative model tests. The following test plan is the minimum obligation.

6.1 Fast-Scan Sensitivity Probe

Status of this test: This is a *sensitivity probe*, not a full model comparison. It tests whether DES Y6 cosmic-shear data responds measurably to EFC-gate (μ, Σ) perturbations around the Planck best-fit Λ CDM cosmology. A negative sensitivity result terminates further investment. A positive sensitivity result motivates the full MCMC model-comparison in §6.2.

This is explicitly not a full $\Delta\chi^2(\text{EFC})$ vs $\Delta\chi^2(\Lambda\text{CDM})$ model comparison. Fixing background cosmology to Λ CDM best-fit gives Λ CDM a structural advantage.

Infrastructure requirement: MGCAMB or MGCLASS with μ - Σ parametrization including k -dependence; tested Cobaya or CosmoSIS likelihood module for DES Y6; collaborator-reviewed pipeline (not AI-generated code).

EFC parametrization (pre-registered):

Scale-dependent form (required, since P3 is k -sensitive):

$$\mu(k, a) = 1 + A_\mu \cdot g(a) \cdot [1 + (k/k_c)^2]^{-1} \quad (1)$$

$$\Sigma(k, a) = 1 + A_\Sigma \cdot g(a) \cdot [1 + (k/k_c)^2]^{-1} \quad (2)$$

$$g(a) = \frac{1}{2}[1 + \tanh((a - a_t)/\delta_a)] \quad (3)$$

Fixed parameters (locked from ledger v3.11 and P3 pre-registration):

- $a_t = 0.7, \delta_a = 0.08$
- $k_c = 0.05 \text{ h/Mpc}$ (matches P3 crossover scale)

Starting value from ledger 2D scan: $A_\mu = -0.06, A_\Sigma = +0.05$.

Parametrization scope (disclosure): The (A_μ, A_Σ) form above is the MGCAMB/MGCLASS-native phenomenological parametrization, chosen for likelihood-pipeline compatibility. It is NOT directly derived from the EFC action. The companion compound paper (P1×P2, DOI 10.6084/m9.figshare.32080059) documents that the action-level parameter K_0 is the primary control parameter for the lensing-sector crossover redshift. A positive fast-scan result indicates phenomenological preference, not direct action-level validation.

Stopping rule: If $\Delta\chi^2$ improvement at best-fit EFC point is < 1 , full MCMC in §6.2 is not justified.

Interpretation of null sensitivity: A null fast-scan result falsifies the specific (A_μ, A_Σ) phenomenological direction tested with locked gate parameters. It does NOT uniquely falsify the EFC action. Alternative (A_μ, A_Σ) regions or gate-parameter configurations consistent with different K_0 -action configurations remain untested until the action-to-phenomenology mapping is constructed.

6.2 Full Model-Comparison MCMC

Status: The *actual* model comparison. Only executed if §6.1 shows measurable sensitivity.

Protocol: Full Cobaya MCMC with both models running on identical DES Y6 likelihood. Λ CDM: vary $\{\Omega_b h^2, \Omega_c h^2, 100\theta_{\text{MC}}, \tau, A_s, n_s\} + \text{nuisance}$. EFC: same cosmological parameters + $\{A_\mu, A_\Sigma\} + \text{nuisance}$. Gate parameters locked.

Outputs: χ^2_{min} both models; ΔAIC , ΔBIC ; full posteriors; consistency check with §6.1 gradient.

Stopping rule: If $|\Delta\text{AIC}| < 2$, models indistinguishable on DES Y6 alone.

6.3 Cross-Pipeline Test

Objective: Determine whether DES Y6 vs KiDS Legacy S_8 -divergence is pipeline-signal or model-signal.

Protocol: Repeat §6.2 against KiDS Legacy with identical parametrization. Compare $\Delta\text{Pipeline}$ under Λ CDM and EFC.

Interpretations:

- $\Delta\text{Pipeline}(\Lambda\text{CDM}) \approx \Delta\text{Pipeline}(\text{EFC})$: divergence is pipeline-driven for both.
- $\Delta\text{Pipeline}(\text{EFC})$ significantly smaller: EFC may be more robust (reportable as robustness signal).
- $\Delta\text{Pipeline}(\text{EFC})$ larger: EFC is *more* sensitive to pipeline (negative robustness signal).

6.4 Symmetric Falsifier

Rule 1 (applied after §6.1): If the fast-scan returns flat χ^2 surface ($\Delta\chi^2$ improvement < 1), further cosmological-sector claims withdrawn. Fast-scan output published as null result regardless.

Rule 2 (applied after §6.2): If $\Delta\text{AIC}(\text{EFC} - \Lambda\text{CDM}) \geq +5$ against DES Y6, with no artifact identified, this note is retracted and EFC's cosmological-sector claims are withdrawn.

Rule 3 (applied after §6.3): If cross-pipeline shows EFC is *more* pipeline-sensitive than Λ CDM at $> 2\sigma$, EFC's claim to robustness under proxy-chain variation is withdrawn.

Commitment: All three rules binding regardless of results. Partial publication of favorable results without corresponding null-result publication constitutes scientific misconduct under this pre-registration.

6.5 Companion Compound Test (P1×P2 Cross-Observable Consistency)

A separate pre-registered consistency test between P1 (sealed lensing crossover at $z \approx 0.44$, DOI 10.6084/m9.figshare.32037990, L2-regime) and P2 (sealed growth prediction $f\sigma_8(z=0.7) = 0.430$, DOI 10.6084/m9.figshare.32013156, L1-regime) is documented in DOI 10.6084/m9.figshare.32080059 (Magnusson 2026). The compound paper establishes empirically that:

1. P1 and P2 reside in disjoint parameter spaces in the current EFC reference implementation.
2. The lensing-sector crossover position is primarily controlled by the action parameter K_0 with amplitude-invariance $\Delta z \approx 0.11$ over a $\times 50$ amplitude rescaling.
3. No unified forward-model mapping $(z_t, n, \mu_0) \rightarrow K_0$ currently exists in the EFC codebase. Constructing such a mapping is a required open task.

6.6 Pre-registration

This note is registered at Figshare under DOI 10.6084/m9.figshare.32080080 (assigned 2026-04-21). The test protocol specifications in §6.1–6.4 are locked and cannot be modified post-data. Any revision requires a new versioned document with new DOI and new content hash.

7 Reproducibility

- All cited papers are listed in §8 with full DOI or arXiv identifiers.
- RCMP source code: github.com/supertedai/EFC/tree/main/rcmp
- EFC parametrization and gate function: `src/rcmp/regime_tagger.py` (already published).
- Test plan scripts: to be committed to github.com/supertedai/EFC/cosmology-forward-model/ prior to fast-scan execution.

8 References

Philosophical / methodological foundations:

- Duhem, P. (1906). *La Théorie Physique: Son Objet et Sa Structure*. Chevalier & Rivière.
- Quine, W.V.O. (1951). Two Dogmas of Empiricism. *Philosophical Review* 60, 20–43.

RCMP and EFC:

- Magnusson, M. (2026). The Regime-Consistent Measurement Principle. [figshare.31222900](https://figshare.com/figure/31222900).
- Magnusson, M. (2026). EFC Action Paper. [figshare.31876324](https://figshare.com/figure/31876324).
- Magnusson, M. (2026). EFC Perturbation Sector (P1). [figshare.32037990](https://figshare.com/figure/32037990).
- Magnusson, M. (2026). EFC Growth Prediction (P2). [figshare.32013156](https://figshare.com/figure/32013156).
- Magnusson, M. (2026). Tech Note II: μ -based σ_8 Suppression. [figshare.31333600](https://figshare.com/figure/31333600).
- Magnusson, M. (2026). P1×P2 Compound Cross-Observable Consistency. [figshare.32080059](https://figshare.com/figure/32080059) (published 2026-04-21).

Cosmological convergence observations:

- Preston, C., Amon, A., Efstathiou, G. (2023). MNRAS 525, 5554.
- Wright et al. (2025). KiDS-Legacy cosmological constraints. A&A (in press).
- Abbott et al. (2026). DES Year 6 Cosmology Results.
- Wang et al. (2025). Did DESI DR2 truly reveal dynamical dark energy? Eur. Phys. J. C.
- Sen et al. (2025). Dynamical dark energy in light of DESI DR2. JHEAp 49, 100428.
- Kamran et al. (2026). Phys. Rev. D 113, 023527.
- Pogosian, L., Silvestri, A. (2016). Constraining Horndeski with Σ and μ . arXiv:1606.05339.
- Amon, A., Efstathiou, G. (2022). A non-linear solution to the S_8 tension. MNRAS 516, 5355.
- 2026 S_8 Tension Review. arXiv:2602.12238.
- DESI Collaboration (2025). DESI DR2 Results II. arXiv:2503.14738.
- ACT Collaboration. DR6 Gravitational Lensing × SDSS BOSS. arXiv:2405.12795.

Historical cosmological proxy-chain debates (context):

- Sandage, A. (1988). Observational tests of world models. *Ann. Rev. Astron. Astrophys.* 26, 561.
- Freedman, W.L. et al. (2001). Final Results from the HST Key Project. *ApJ* 553, 47.

9 Release Conditions

This draft is released under DOI 10.6084/m9.figshare.32080080 (2026-04-21). The published v0.2 status reflects:

1. §6.1 Fast-Scan: not yet completed. Publication as null result if and when executed.
2. §6.1 result published regardless of outcome. If null result, it is published as null result; the observation in §§2–4 stands.
3. §6.2 Full MCMC: conditional on §6.1 positive sensitivity.
4. §6.2 preliminary result required before future companion papers cite this as “EFC cosmological test.”
5. Ledger updated incorporating §6.1 and §6.2 outputs under RCMP validation-checklist format, regardless of outcome.
6. Compound paper (DOI 10.6084/m9.figshare.32080059) released as companion 2026-04-21.

Stopping rules (binding):

If §6.1 shows no sensitivity ($\Delta\chi^2$ improvement < 1 at best-fit EFC point): the null result is published; the methodological convergence observation in §§2–4 remains valid independently.

If §6.2 yields $\Delta\text{AIC} \geq +5$ against EFC: retraction and EFC-cosmology-claim-withdrawal is executed per §6.4 Rule 2.